

BATTERY SAFETY CHARACTERIZATION

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Battery Safety Characterization

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I Sammanfattning

Denna avhandling undersöker beteendet hos termisk rusning i cylindriska NMC811 litiumjonceller genom att analysera data erhållna via Accelerating Rate Calorimetry (ARC). Studien fokuserar på att analysera temperatur, tryck och massförlustandel vid olika laddningsgrader (SOC). Viktiga resultat inkluderar identifieringen av termoelementplacering som kritisk för noggrann temperaturdetektering, inverkan av SOC på intensiteten hos termisk rusning, och korrelationen mellan ventilationstryck och massförlust. Resultaten belyser vikten av flerparameteranalys vid karakterisering av termiska händelser och ger insikter för att förbättra batterisäkerhetsmekanismer. Forskningen bidrar till förståelsen av felmekanismer i högenergiceller och informerar framtida batteridesign och testprotokoll.

Nyckelord: Litiumjonbatterier, NMC811, Termisk rusning (TR), Batterisäkerhet, Accelerating Rate Calorimetry (ARC), Laddningsgrad (SOC), Termisk hantering, Ventilationstryck, Temperatur, Massförlust

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II ABSTRACT

This thesis investigates the thermal runaway behavior of cylindrical NMC811 lithium-ion battery cells via analyzing the data obtained via Accelerating Rate Calorimetry (ARC). The study focuses on analyzing temperature, pressure, and mass loss fraction across various states of charge (SOC). Key findings include the identification of thermocouple placement as critical for accurate temperature detection, the influence of SOC on thermal runaway intensity, and the correlation between venting pressure and mass loss. The results highlight the importance of multi-parameter analysis in characterizing thermal events and provide insights for improving battery safety mechanisms. The research contributes to the understanding of failure mechanisms in high-energy cells and informs future battery design and testing protocols.

Key words: Lithium-ion batteries, NMC811, Thermal Runaway, Battery Safey, Accelerating Rate Calorimetry, State of Charge (SOC), Thermal Management, Venting Pressure, Temperature, Mass Loss.

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IV NOMENCLATURE

Here are the Notations and Abbreviations that are used in this Master thesis

Abbreviations

ARC	Accelerated Rate Calorimetry
HWS	Heat-Wait-Seek Method
MLF	Mass Loss Fraction
NMC811	Nickel-Manganese-Cobalt-811
PPE	Polyphenylene Ether
TR	Thermal Runaway

V TABLE OF CONTENTS

Contents

I SAMMANFATTNING 1

II ABSTRACT 2

III ACKNOWLEDGEMENTS. 3

IV NOMENCLATURE 4

V TABLE OF CONTENTS 5

VI LIST OF FIGURES 7

VII LIST OF TABLES 7

1. INTRODUCTION 9

1.1 Motivation And Objective 10
1.2 Methodology 10
1.3 Limitations 11

2. THEORETICAL BACKGROUND 11

2.1 NMC811 Cell Chemistry and Safety Studies 11
2.2 Thermal Runaway Mechanism in NMC811 at Cell level 11
2.2.1. SEI Decomposition (70–120°C) 12
2.2.2. Anode + Electrolyte Reaction (85–200°C) 12
2.2.3. PE Separator Melting (~130°C) 12
2.2.4. Ceramic Coating Collapse (~200°C) 12
2.2.5. NMC811 Cathode + Electrolyte Reaction (175–312°C) 12
2.2.6. Electrolyte Decomposition (180–300°C) 13
2.2.7. Internal Short Circuit 13
2.2.8. Electrolyte Ignition and Burning (>300°C) 13
2.2.9. Full Thermal Runaway (>500°C) 13
2.3 Various Abuse Testing methods 14
2.4 ARC Testing 14

2.4.1 Heat-Wait-Seek (HWS) Method 15
2.4.2 Multi-Parameter Measurement Techniques 15
2.4.3 Stages of Thermal Runaway for ARC testing 15

3 IMPLEMENTATION 16

3.1 Arc testing setup and parameters 16
3.2 Criteria for Parameter Analysis 19
3.2.1 Temperature 19
3.2.2 Pressure 19
3.2.3 Mass Loss Fraction 19

4 RESULTS AND DISCUSSION 20

4.1 Temperature Analysis 20
4.2 Temperature Rate analysis 26
4.3 Maximum Temperature Analysis 28
4.4 Multiple Parameter analysis 30
4.5 Mass Loss Fraction Analysis 31
4.6 Venting Pressure Analysis 32

5. CONCLUSIONS 34

6 RECOMMENDATIONS AND FUTURE WORK 35

6.1 Recommendations 35
6.2 Future Work 35

7 REFERENCES 36

VI LIST OF FIGURES

Figure 1: Thermal Runaway Mechanism 13
Figure 2: HWS Method flowchart 15
Figure 3: ARC testing Setup 17
Figure 4: Schematic Diagram of the thermocouple attachment location 18
Figure 5: Test 12; Temperature and Pressure vs Time 20
Figure 6: Section 1 for figure 5 21
Figure 7: Section 2 for Ilustración 5 21
Figure 8: Section 3 for Ilustración 5 22
Figure 9: Test 2; Temperature and Pressure vs Time 22
Figure 10: Test 1; Temperature and Pressure vs Time 22
Figure 11: Test 9; Temperature and Pressure vs Time 23
Figure 12: Test 4; Temperature and Pressure vs Time 23
Figure 13: Test 3; Temperature and Pressure vs Time 23
Figure 14: Test 6; Temperature and Pressure vs Time 24
Figure 15 Test 5; Temperature and Pressure vs Time 24
Figure 16: Test 11; Temperature and Pressure vs Time 24
Figure 17: Test 7; Temperature and Pressure vs Time 24
Figure 18: Test 10; Temperature and Pressure vs Time 25
Figure 19: Test 8; Temperature and Pressure vs Time 25
Figure 20: Temperature vs Time for different SOC 25
Figure 21: Test 1; Temperature Rate vs Time 26
Figure 22: Test 2; Temperature Rate vs Time 26
Figure 23: Test 12; Temperature Rate vs Time 26
Figure 24: Test 9; Temperature Rate vs Time 26
Figure 25: Test 4; Temperature Rate vs Time 27
Figure 26: Test 3; Temperature Rate vs Time 27
Figure 27: Test 6; Temperature Rate vs Time 27
Figure 28: Test 5; Temperature Rate vs Time 27
Figure 29: Max Temperature vs SOC 29
Figure 30: Multiple Parameter analysis (Temperature, Pressure and Mass Loss Fraction) 30
Figure 31: MLF vs Max Pressure at different SOC 31
Figure 32: MLF vs Max Temperature for different SOC for Vent location 32
Figure 33: MLF vs Max Temperature for different SOC for Bottom location 32
Figure 34: MLF vs Max Temperature for different SOC for Mid location 32
Figure 35: MLF vs Max Temperature for different SOC for Top location 32
Figure 36: Venting Pressure vs Max Temperature at different SOC levels 33

VII LIST OF TABLES

Table 1: Maximum temperature for all the channels at their respective SOC level 28
Table 2: Self heating, Venting and Initiation temperature for all Channels at their respective SOC level 29
Table 3: Max Temperature at Channel 3 in comparison the Max Pressure and Mass loss Fraction for all the SOC levels 30
Table 4: Venting Pressure and Vent Max temperatures at various SOC 33
Table 5: Range of Temperature and pressure over SOC level 34

1. INTRODUCTION

The global transition toward electrification, particularly in transportation and renewable energy storage, has intensified the demand for high-performance lithium-ion batteries (LIBs). Among the various chemistries available, **Nickel Manganese Cobalt Oxide (NMC)**—especially the **NMC811** variant (comprising 80% nickel, 10% manganese, and 10% cobalt as the transition metal)—has emerged as one of the leading candidates due to its high energy density, improved capacity retention in comparison to traditional options such as LFP, and cost-effectiveness[1]. NMC811 is more widely adopted than alternatives like NMC 622 due to its higher energy density, and it is favored over NMC 90:5:5 for offering better thermal and structural stability[2]. However, these advantages come with significant safety challenges, particularly under abusive conditions that can lead to **thermal runaway**.

Thermal runaway is a critical failure mode in lithium-ion batteries, characterized by a self-sustaining exothermic reaction that can result in rapid temperature escalation, gas venting, fire, or even explosion[3]. Tr is triggered by internal short circuits, mechanical damage, overcharging, or external heating[4]. Once initiated, the process is difficult to control and poses serious risks to users, infrastructure, and the environment. Understanding the mechanisms and thresholds of thermal runaway is therefore essential for developing safer battery systems.

To investigate thermal runaway behavior, researchers often employ **Accelerating Rate Calorimetry (ARC)**—an accurate and versatile technique that allows for the detection of exothermic reactions under adiabatic conditions[5]. ARC testing provides critical insights into the onset temperature of thermal events, heat generation rates, and the total energy released during decomposition. This method is particularly valuable for characterizing the thermal stability of battery cells and evaluating the effectiveness of safety features such as pressure relief vents and shutdown separators .

In the context of **cylindrical NMC811 cells**, safety characterization is especially important due to their widespread application in electric vehicles and stationary storage systems. These cells, while efficient in energy storage, are more prone to thermal instability due to the high nickel content, which increases reactivity and reduces thermal stability[6]. Previous studies have shown that NMC811 cells exhibit lower onset temperatures for thermal runaway compared to other chemistries like LFP (Lithium Iron Phosphate), and their failure modes can vary significantly depending on cell design, state of charge, and external conditions[7].

Several researchers have explored the thermal behavior of NMC cells using ARC and other calorimetric methods. For instance, studies have reported that thermal runaway in NMC811 cells can initiate at temperatures as low as 180–200°C, with peak temperatures exceeding 600°C. These findings underscore the importance of precise thermal management and robust safety mechanisms in battery pack design[8].

This thesis builds upon such foundational work by in-depth analysis of empirical data, provided with experimental results for the thermal runaway characteristics of cylindrical NMC811 cells. Using ARC testing and complementary diagnostic tools, the study aims to provide a deeper understanding of the thermal behavior, failure mechanisms, and safety implications of these high-energy cells.

1.1 Motivation And Objective

As the adoption of lithium-ion batteries accelerates across industries, the demand for safer, more reliable energy storage system becomes increasingly urgent. While NMC811 cells offer superior energy density and cost efficiency, their high nickel content introduces significant thermal and chemical instability. This trade-off between performance and safety presents a critical engineering challenge. Recent high-profile incidents involving battery fires in electric vehicles and consumer electronics have underscored the need for rigorous safety assessments. Understanding the thermal behavior of these cells under abuse conditions is not only essential for preventing severe failures but also for informing the design of next-generation battery management systems and safety protocols.

The primary objective of this thesis is to conduct a comprehensive safety characterization of cylindrical NMC811 lithium-ion cells, with a specific focus on their response to thermal abuse.

The study aims to:

- Quantify the thermal thresholds associated with thermal runaway.
- Understand the optimum position for temperature measurement on a cylindrical cell in abuse testing
- Evaluate the effectiveness of safety features such as venting mechanisms.
- Identify and analyze various parameters for thermal runaway

By achieving these goals, the research seeks to contribute actionable insights for battery manufacturers, system integrators, and safety regulators.

1.2 Methodology

The research methodology is structured into several key phases:

1 Experimental Setup: A controlled laboratory environment is established using an Accelerating Rate Calorimeter (ARC) to simulate thermal abuse scenarios. The setup includes multiple high-resolution thermal sensor and pressure sensors. Tests are carried out by test engineers.
2 Sample Preparation: Commercial 4695 (cylindrical format with 46 mm diameter and 95 height) NMC811 cells from are selected at various SOC levels (comment on cycling) and conditioned to a consistent state of charge, which is 07%, 030%, 050%, 090%, 100%. Cells are inspected for uniformity and integrity before testing.
3 Thermal Abuse Protocol: Cells are subjected to a stepwise heating protocol to induce thermal runaway. The ARC system automatically adjusts heating rates based on exothermic activity, ensuring adiabatic conditions.
4 Data Collection: Time series data on temperature, pressure, and gas evolution are recorded.
5 Post-Test Analysis: After testing, cells are visually inspected for structural damage. Data is analyzed to determine critical safety parameters such as onset temperature, maximum temperature, and time-to-runaway.
6 Data Analysis : Various parameters are analyzed using python to conclude from the data

1.3 Limitations

While the study is designed to provide robust insights, several limitations must be acknowledged:

- Single-Cell Focus: The experiments are limited to individual cells and do not account for interactions within multi-cell modules or packs.
- Controlled Conditions: Laboratory conditions may not fully replicate real-world abuse scenarios such as mechanical impact or electrical overcharge.
- Instrumentation Constraints: Sensor placement and resolution may limit the ability to capture internal dynamics with high precision.
- Scope of Chemistry: The study is limited to NMC811 cells and does not generalize for other cell chemistries such as Lithium Iron Phosphate or Nickel Cadmium.

2. THEORETICAL BACKGROUND

2.1 NMC811 Cell Chemistry and Safety Studies

Nickel Manganese Cobalt Oxide (NMC) cathodes are widely used in lithium-ion batteries due to their high energy density and relatively balanced performance. Among these, **NMC811** ($LiNi_{0.8}Mn_{0.1}Co_{0.1}O_2$) has gained prominence for its high nickel content, which enhances capacity and reduces reliance on cobalt. However, this comes at the cost of **reduced thermal stability** and increased reactivity under abuse conditions[7].

The high nickel content in NMC811 leads to greater oxygen release during decomposition, which can fuel combustion reactions during thermal runaway. Studies have shown that NMC811 cells exhibit **lower onset temperatures** for thermal runaway and **higher heat release rates** compared to NMC 622 or LFP cells[9]. Researchers have explored various mitigation strategies, including:

Surface coatings (e.g., Al_2O_3, ZrO_2) to stabilize the cathode surface,

Electrolyte additives to suppress gas generation,

Advanced separators with ceramic coatings to delay internal short circuits.

Despite these improvements, the inherent instability of NMC811 under thermal stress remains a significant safety concern, especially in high-capacity applications like electric vehicles and grid storage[4].

2.2 Thermal Runaway Mechanism in NMC811 at Cell level

External heating is a type of thermal abuse which involves heating of a battery cell beyond its safe operating temperature. This can occur due to environmental exposure, malfunctioning thermal

management systems, or adjacent cell failures. In the study by Yan et al. thermal abuse was simulated using both internal and external heaters to initiate failure in 18650 NMC811 cells at various SOCs [10].

2**.2.1**. SEI Decomposition (110–120°C)

The Solid Electrolyte Interphase (SEI i.e. interaction between anode and electrolyte) on the anode begins to decompose at relatively low temperatures. This layer, formed during initial battery cycling, is thermally unstable and breaks down to release gases such as CO_2, C_2H_4, and Li_2CO_3.

Reaction:

$(CH_2OCO_2Li)_2 \rightarrow Li_2CO_3 + C_2H_4 + CO_2 + 12O_2 (CH_2OCO_2Li)_2 \rightarrow Li_2CO_3 + C2H_4 + CO_2 + 21O_2$

This decomposition is exothermic and contributes to the initial temperature rise[5].

2**.2.2**. Anode + Electrolyte Reaction (85–200°C)

As the temperature increases, intercalated lithium in the anode reacts with the organic electrolyte solvents (e.g., EC, PC, DEC), producing CO, CH_4, C_2H_6, and C_3H_6.

Example reactions:

$C_3H_4O3 + 2Li \rightarrow (CH2OLi)_2 + CO$
$C_3H_4O_3 + 2Li \rightarrow (CH_2OLi)_2 + CO$
$C_5H_{10}O_3 + 2Li \rightarrow (CH_3CH_2OLi)_2 + CO$
$C_5H_{10}O_3 + 2Li \rightarrow (CH_3CH_2OLi)_2 + CO$

These reactions are highly exothermic and generate flammable gases[11].

2**.2.3**. PE Separator Melting (~130°C)

The polyethylene (PE) separator melts, compromising the physical barrier between the anode and cathode. This increases the risk of internal short circuits[11], [12].

Melting point of PE: ~130°C

Consequence: Loss of mechanical integrity and potential for direct electrode contact.

2**.2.4**. Ceramic Coating Collapse (~200°C)

If present, ceramic coatings on separators or electrodes begin to degrade. These coatings are designed to delay thermal propagation, but at elevated temperatures, they lose structural integrity. Collapse leads to increased ionic conductivity and potential shorting[13].

2**.2.5**. NMC811 Cathode + Electrolyte Reaction (175–312°C)

The NMC811 cathode undergoes decomposition, releasing oxygen and reacting with the electrolyte. This is the most critical stage in thermal runaway.

Reactions:

$LiNiO_2 \rightarrow NiO + O_2 LiNiO_2 \rightarrow NiO + O_2$
$LiCoO_2 \rightarrow CoO + O_2 LiCoO_2 \rightarrow CoO + O_2$

The released oxygen supports combustion of the electrolyte and gases[13].

2**.2.6**. Electrolyte Decomposition (180–300°C)

The organic solvents in the electrolyte decompose further, producing CO, CO_2, H_2, and hydrocarbons. These reactions are catalyzed by the high temperatures and oxygen released from the cathode[14].

Example: $C_3H_4O_3 + 2Li \rightarrow Li_2CO_3 + C_2H_4 C_3H_4O_3 + 2Li \rightarrow Li_2CO_3 + C_2H_4$

2**.2.7**. Internal Short Circuit

Due to separator failure and electrode deformation, internal short circuits occur. This leads to localized heating, which can rapidly escalate the temperature and trigger full thermal runaway. Short circuits are often the final trigger before runaway[14].

2**.2.8**. Electrolyte Ignition and Burning (>300°C)

The flammable gases and decomposed electrolyte ignite in the presence of oxygen, leading to combustion. This stage is marked by flames, pressure spikes, and ejection of hot particles. Flame jets observed at high SOCs[10].

Gases involved: H_2, CO, CH_4, C_2H_4, C_3H_6

2**.2.9**. Full Thermal Runaway (>500°C)

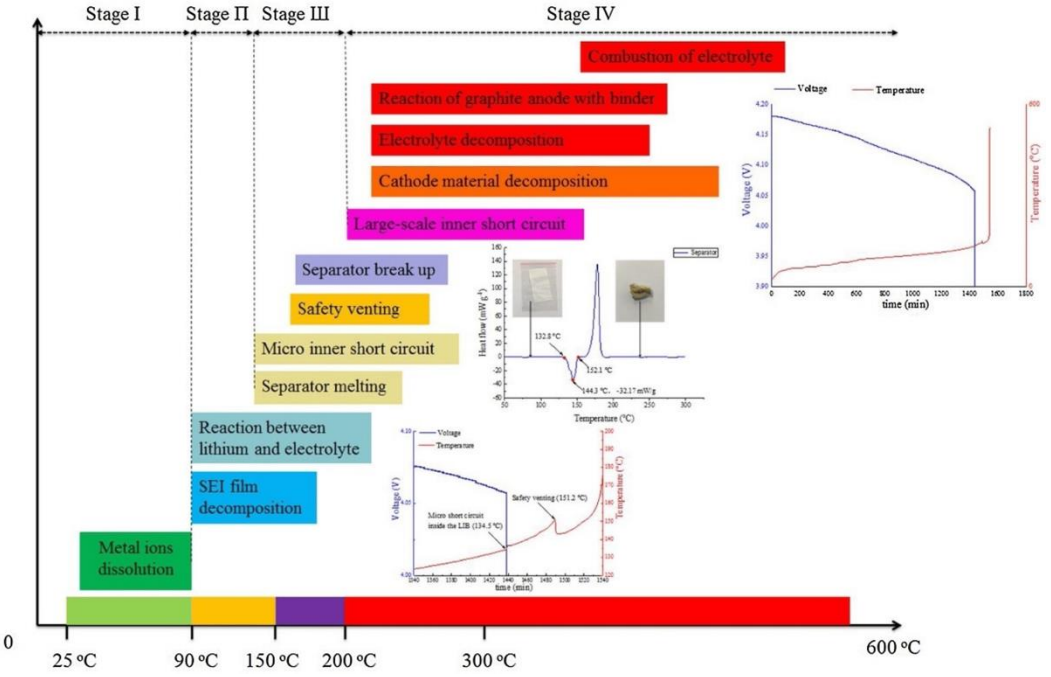


Figure 1: Thermal Runaway Mechanism

[13]

Thermal runaway becomes self-sustaining, with temperatures exceeding 800–1000°C. The cell may rupture or explode, releasing toxic gases such as HF, HCN, and $Ni(CO)_4$. This is a highly toxic gas detected in the study's pressure vessel tests.[15]

Nickel carbonyl formation:
$Ni+4CO \rightarrow Ni(CO)$

$4Ni+4CO \rightarrow Ni(CO)_4$

2.3 Various Abuse Testing methods

To evaluate battery safety, cells are subjected to **abuse tests** that simulate real-world failure scenarios. These tests are standardized in protocols such as **UL 1642**, **IEC 62133**, and **UN 38.3**. The main categories include:

Thermal Abuse:

Cells are heated externally (e.g., in an oven or ARC) to simulate overheating. This is the most direct method to induce thermal runaway.

Electrical Abuse:

Overcharge: Forces lithium plating and gas generation.
Over-discharge: Can lead to copper dissolution and internal shorts.
Short Circuit: Rapid current flow generates intense localized heating.

Mechanical Abuse:

Nail Penetration: Simulates internal shorting due to physical damage.
Crush and Impact: Tests structural integrity under compression or collision.

Environmental Abuse:

Includes vibration, humidity, and altitude tests to assess robustness in transport and operation.
Each test provides insights into different failure mechanisms and helps validate the effectiveness of safety features like **current interrupt devices**, **pressure relief vents**, and **flame-retardant electrolytes**[9].

2.4 ARC Testing

The Accelerating Rate Calorimeter (ARC) is a critical tool for evaluating the thermal stability and runaway behavior of lithium-ion cells under adiabatic conditions. It enables researchers to simulate worst-case thermal abuse scenarios by monitoring self-heating reactions without external heat loss. This method is particularly effective for identifying the onset of thermal runaway and quantifying the heat and gas generation associated with it[16].

In the study by Lei et al. (2017), commercial 18650 cells with various cathode chemistries—including NMC, LFP, and LMO—were tested using an es-ARC system. The focus was on understanding the thermal behavior of fully charged (100% SOC) cells under controlled heating conditions[17].

2.4.1 Heat-Wait-Seek (HWS) Method

The standard Heat-Wait-Seek (HWS) protocol was employed to detect the onset of exothermic reactions[13]:

Heat Mode: The cell is heated in 5°C increments starting from 25°C.
Wait Mode: A 15-minute pause allows thermal equilibrium.
Seek Mode: The system monitors the self-heating rate. If it exceeds a threshold (e.g., 0.02°C/min), the ARC switches to Exotherm Mode, simulating adiabatic conditions.

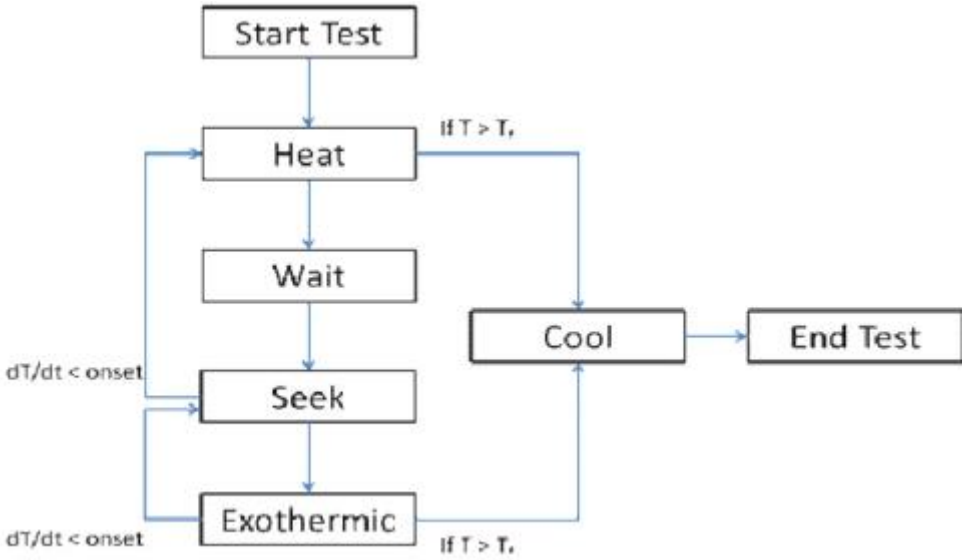


Figure 2: HWS Method flowchart

This method allows precise identification of the temperature at which self-heating begins and tracks the progression of thermal runaway.

2.4.2 Multi-Parameter Measurement Techniques

Four experimental methods were used to enhance the understanding of thermal runaway:

Temperature-Only Measurement (OT): Captures surface temperature and heating rate.
External Pressure Measurement (EP): Detects gas venting by monitoring pressure in a sealed chamber.
Internal Pressure Measurement (IP): Measures pressure inside the cell using a capillary inserted through the anode tab.
Ramp Heating Method: Continuously heats the cell at 5°C/min to simulate hot-box conditions.
These methods provided complementary insights into the thermal and pressure dynamics during runaway.

2.4.3 Stages of Thermal Runaway for ARC testing

ARC testing revealed three distinct stages of thermal runaway[16]:

Stage 1 (~90°C): Decomposition of the SEI layer, initiating gas generation. Also known as self-heating.
Stage 2 (~110–200°C): Exothermic reactions between lithium and electrolyte, leading to pressure buildup. Also known as Venting.
Stage 3 (>200°C): Cathode–electrolyte reactions and oxygen release, resulting in rapid temperature rise and cell venting. Also known as rapid thermal runaway.

3 IMPLEMENTATION

This study utilizes experimental data obtained from **Accelerating Rate Calorimetry (ARC)** tests conducted on lithium-ion battery cells to investigate thermal runaway behavior. The ARC method is particularly suited for detecting self-heating reactions under adiabatic conditions, making it a critical tool for evaluating battery safety.

3.1 ARC testing setup and parameters

The ARC tests were performed using the **4695** lithium-ion cell type, with tests conducted in an **air atmosphere**. The cells were tested across a range of **state-of-charge (SOC)** levels, including 0.07, 0.3, 0.5, 0.9, and 1.0 (100%). A total of **12 tests** were conducted, with **SOC = 1.0** being the most frequently tested condition (4 tests), followed by two tests each at SOC levels 0.9, 0.5, 0.3, and 0.07.

The ARC testing followed a **Heat-Wait-Seek (HWS)** protocol, which is standard for detecting the onset of exothermic reactions:

- **Initial temperature**: ~25 °C
- **Heating phase**: 5 °C increase over 10 minutes (0.5 °C/min)
- **Wait phase**: 30 minutes (extended to 60 minutes for PPE cells)
- **Seek phase**: 20 minutes to detect self-heating
- **Exothermal mode**: Triggered if the self-heating rate exceeds **0.02 °C/min** (defined as **T1**)
- **End temperature**: Typically 275 °C, but adjusted to 300 °C or 200 °C in specific tests to protect the facility

The **temperature step** between each HWS cycle was 5 °C. The test was terminated either upon reaching the end temperature or upon detection of a significant exothermic event.

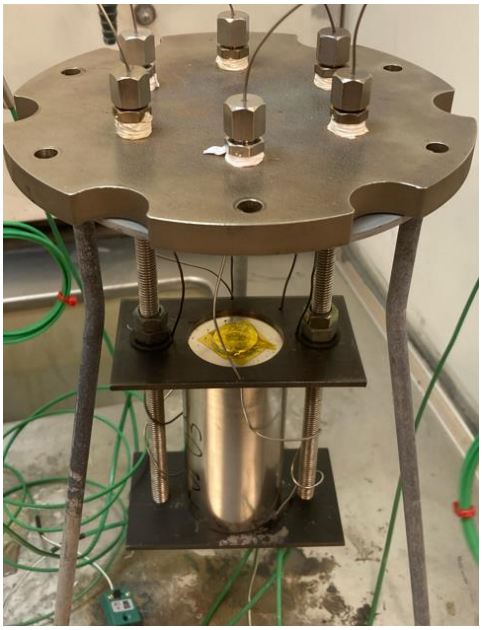


Figure 3: ARC testing Setup

Each ARC test generated a rich dataset comprising:

- **Temperature (°C)**: Internal and external cell temperatures
- **Pressure (bar)**: Measured inside the calorimeter chamber
- **Time (min)**: Timestamp for each data point
- **Self-heating rate (°C/min)**: Derived from temperature derivatives
- **Mass measurements (g)**: Start and end mass of the cell to assess material loss

The data was logged at a frequency of **one sample every 20 seconds** during the HWS phase. The raw data was exported in .DAT and .EXO formats:

- .DAT files contain all measurement points
- .EXO files capture data during exothermic events

Additionally, external pressure sensors were used, with calibration based on current (4–20 mA corresponding to 1.01–40 bar).

The table below summarizes key parameters from the ARC tests:

SOC	***Number of Tests***	***Typical Mass Loss (g)***	***Notable Observations***
100	4	~285–312	Pressure issues, fixture testing
090	2	~271–274	Sensor optimization
050	2	Varies (some missing data)	Thermocouple changes, jelly roll ejection
030	2	~27–28	Transport-charged cells
007	2	~28–30	Thermocouple failures

Mass loss was calculated as the difference between the initial and final cell mass. For example, in one test at SOC 1.0, the mass dropped from **427.7 g to 141.78 g**, indicating a loss of **285.92 g**, likely due to venting and ejection of internal materials.

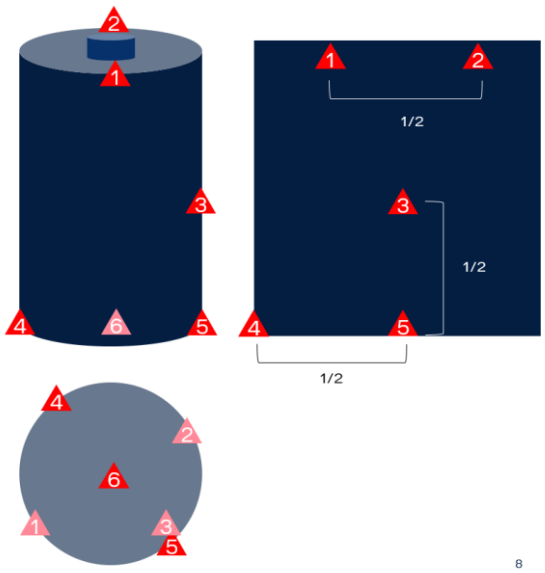


Figure 4: Schematic Diagram of the thermocouple attachment location

One unique feature of this setup was the amount of thermocouples used for detecting temperature. This test employed 6 thermocouples at different cell locations making it first of its kind with precise temperature profile from all over the cell.

The attached thermocouples are then defined on the basis of their locations, TOP is used to refer to thermocouple 1 and 2. Mid is used to refer to thermocouple 3. Bottom is used to refer to thermocouple 4 and 5. Vent is used to refer to thermocouple 6. It is important to make a sharp distinction because certain locations may show different trends.

Key Themes and Recurring Issues

1. **Sensor Failures and Redesigns**

    - Thermocouple breakage and insulation failure were recurring problems.
    - These led to false readings and software malfunctions, prompting hardware redesigns (e.g., brass cylinder + Swagelok fittings).

2. **Pressure Monitoring Challenges**

    - Clogged or leaking pressure lines were observed in multiple tests.
    - These issues compromised the ability to detect gas release and required hardware upgrades.

3. **Cell Integrity and Ejection Events**

- At least one test (SOC 0.5) resulted in jelly roll ejection, indicating a severe rupture.
- This highlights the violent nature of thermal runaway and the need for robust containment.

4. **Impact of Cell Conditioning**

    - Tests using uncycled or transport-charged cells may have introduced variability.
    - Proper preconditioning (e.g., cycling) is essential for consistent thermal behavior.

5. **Instrumentation Evolution**

    - The testing campaign involved iterative improvements to the setup, including thermocouple types, insertion methods, and pressure line configurations.
    - These adaptations reflect a learning curve and a commitment to improving data quality and safety.

3.2 Criteria for Parameter Analysis

A cell's performance usually depends on multiple parameters, however, for this study, the following parameters were selected for the analysis during ARC testing of NMC – 811 cells:

- Temperature
- Pressure
- Mass loss fraction

These parameters were selected on the basis of their critical roles in characterizing the onset, progression and severity of thermal runaway events in Li-ion batteries.

3.2.1 Temperature

Temperature is a primary indicator of thermal activity within the cell. Onset of self-heating and the maximum thermal intensity during thermal runaway are directly provided by temperature. Precise thermal analysis is essential for understanding heat generation and dissipation in Li-ion batteries, which is critical for their safety and performance.

3.2.2 Pressure

Change is pressure is a direct consequence of exothermic reactions and the gas release during those reactions. Many reactions that lead to thermal runaway are a result of combination of temperature and pressure rise. Hence pressure is an important metrics to be analyzed.

3.2.3 Mass Loss Fraction

Mass loss rate is an important indication of thermal runaway severity and it often correlated with volume and composition of flammable gases released. This parameter also helps also helps understand an peculiar relationship with temperature which will be explored further in results.

All the above identified parameters are then compared using Python which provides a wide range of options to carry out data analysis

4 RESULTS AND DISCUSSION

In the results chapter the results that are obtained with the process/methods described in the previous chapter are compiled, and analyzed and compared with the existing knowledge and/or theory presented in the frame of reference chapter.

Accelerated Rate Calorimetry was employed to simulate thermal runaway in NMC811 cylindrical cells. The tests provided insights into temperature profiles across various cell location, pressure changes and mass loss during TR events across various States of Charge (SOC).
Results are further divided into various sections as per the parameter analyzed.

4.1 Temperature Analysis

Analysis starts with intra-cell comparison at a single SOC level. Temperatures from each thermocouple are compared with each other for same test to identify and analyze importance of location for temperature detection.

SOC 100%:

Four tests; test number 1, 2, 9, 12 were carries out with SOC level 100%. Data from all the locations (thermocouples) is compared against each other to verify them.

Test number 12: SOC 100%

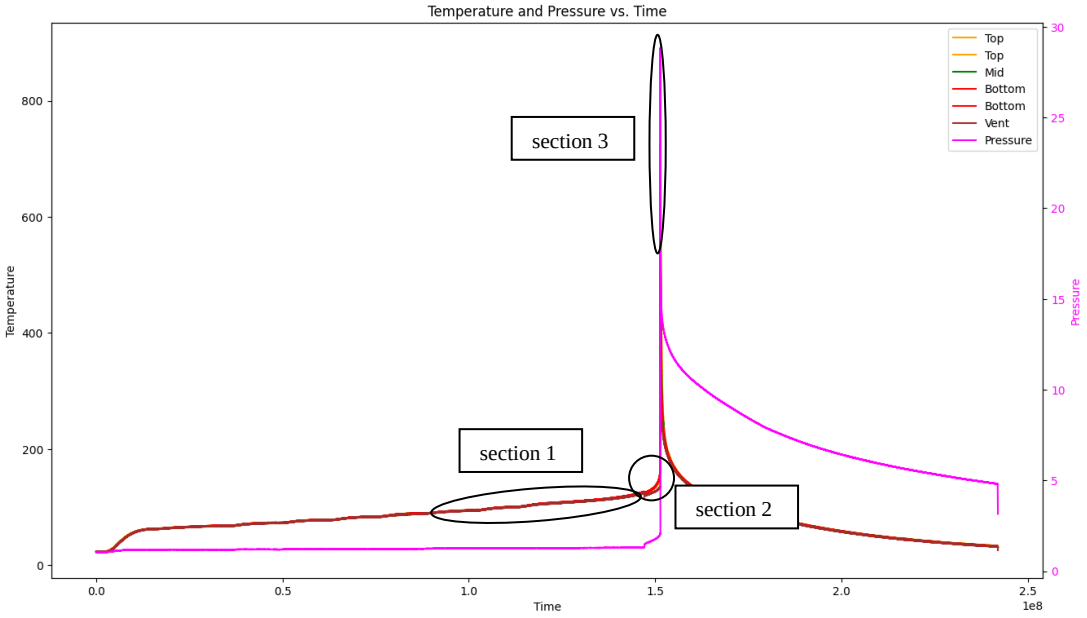


Figure 5: Test 12; Temperature and Pressure vs Time

The curve here appears to be a single line for temperature and a single line for pressure, however, all the data from different thermocouple is so similar over a large range, plot finds it difficult to understand the differences. But if a specific region is zoomed in, differences in temperature and easy to spot.

Throughout the initial heating and self-heating stage, it can be observed that the temperature in different thermocouples varied from not more than 0.5 deg C. This proved the accuracy for thermocouple and all the temperature variations observed later will mostly be due to TR and not a technical error. This part of curve is named section 1 just for the sake of understanding temperature profile in that section.
Section 1:

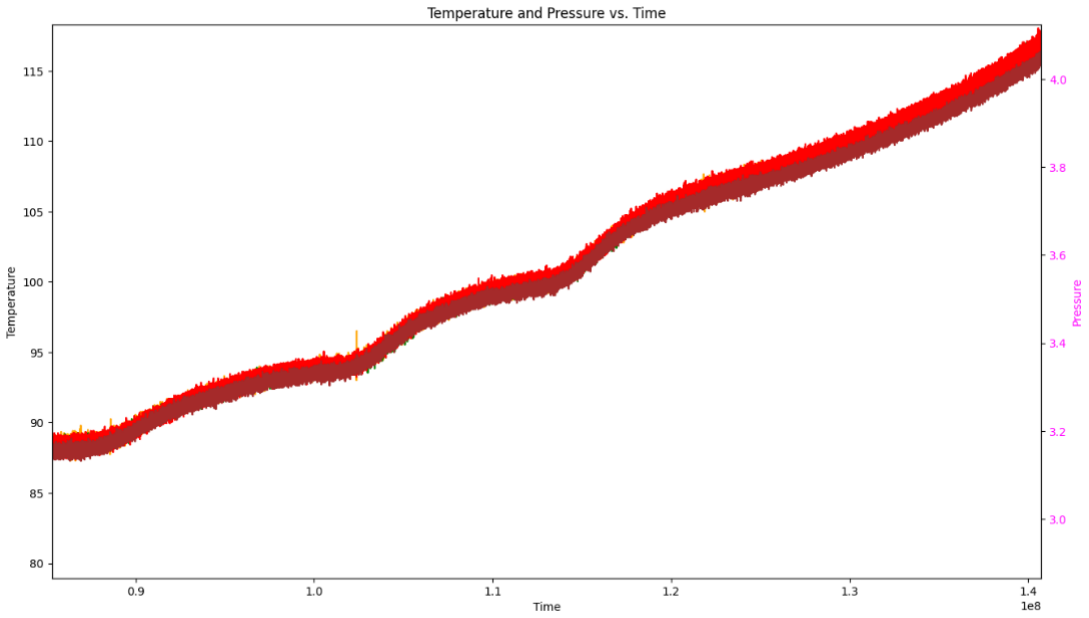


Figure 6: Section 1 for figure 5

However, the nature of plot changes in its TR course. The curves begin to separate as TR initiates and are completely distinct by the time TR is at its maximum intensity. While the trends might not be consistent from TR in cells at different SOC level, it is clear from all the tests that different locations have different behavior during TR. This part of curve, from venting to thermal runaway initial, is named as section 2. Next critical part of the curve is section 3, where TR reaches its maximum intensity.
Section 2:

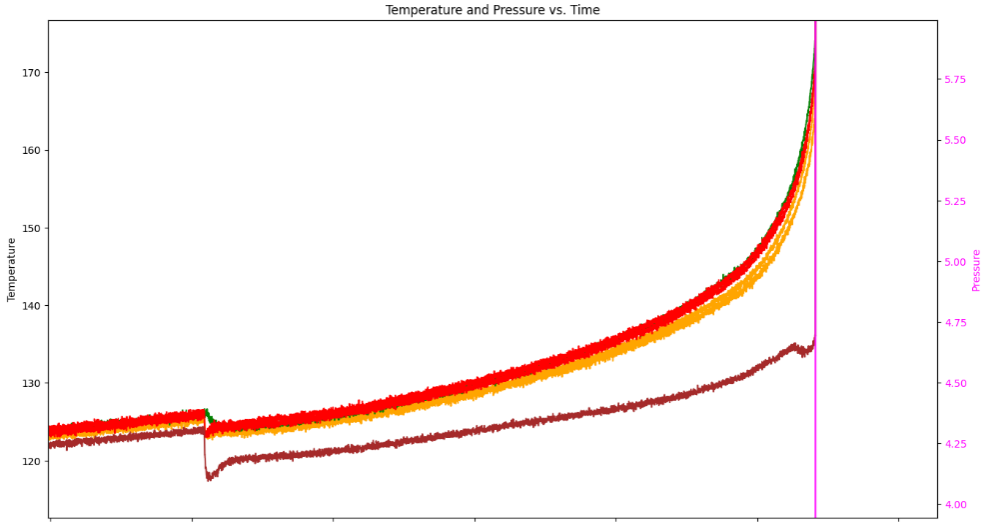


Figure 7: Section 2 for Ilustración 5

Section 3:

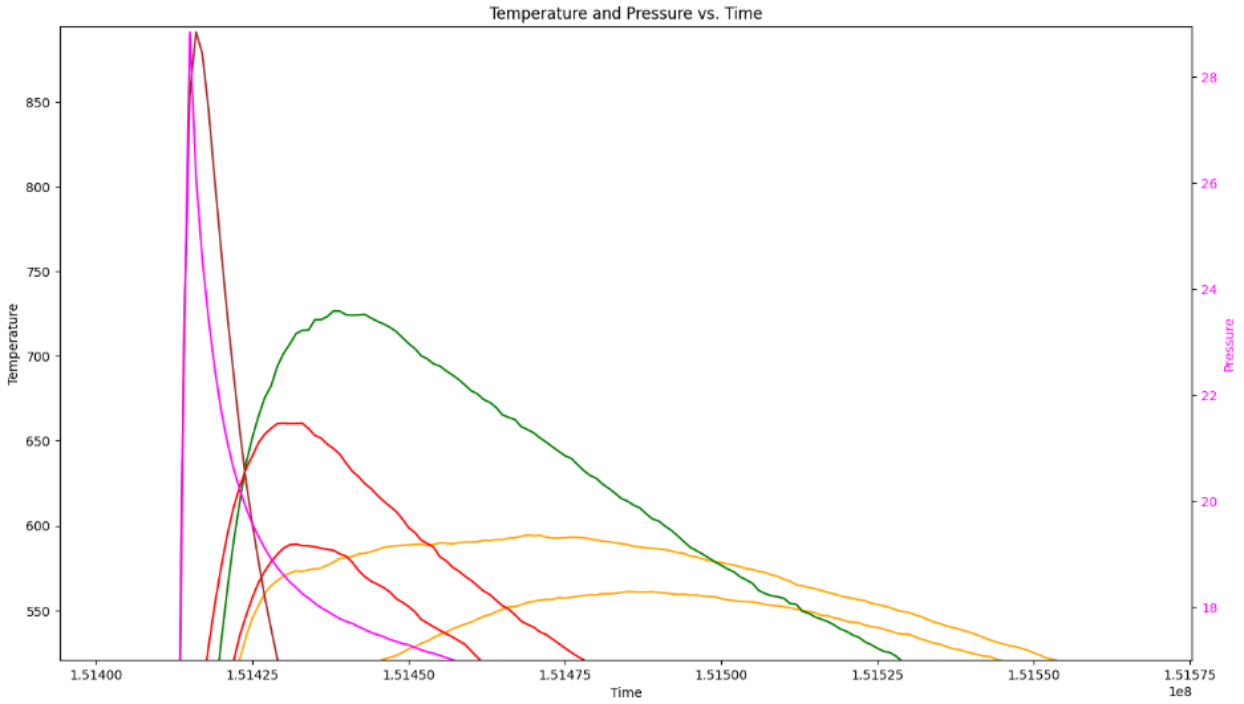


Figure 8: Section 3 for Ilustración 5

Section 3 clearly proves the necessity to study various sections of a cell at the same time because of such distinct variation in temperature at the peak of TR. Other tests with SOC 100% show similar trend.

Other tests:

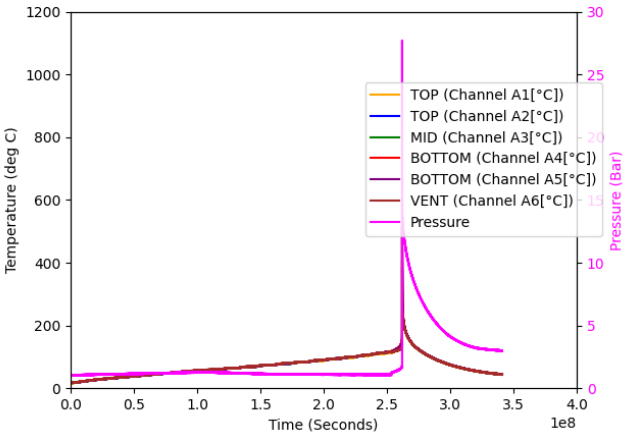


Figure 10: Test 1; Temperature and Pressure vs Time

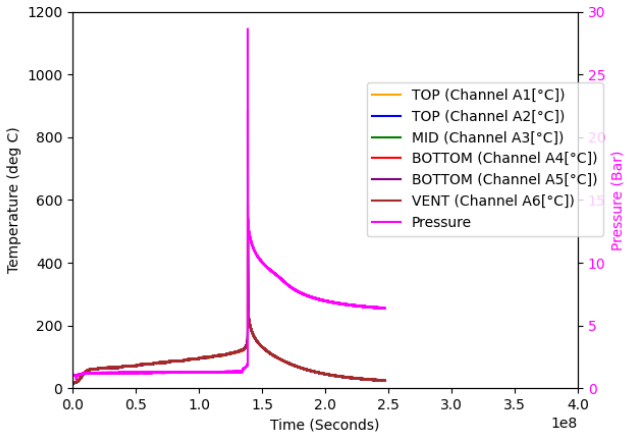


Figure 9: Test 2; Temperature and Pressure vs Time

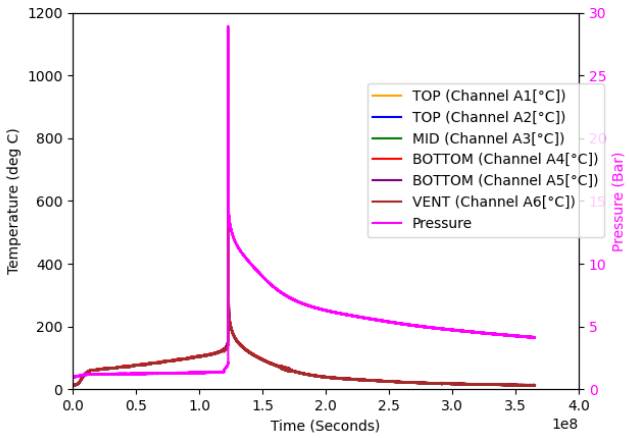


Figure 11: Test 9; Temperature and Pressure vs Time

SOC 90%:

Two tests; test number 3, 4 were carries out with SOC level 90%. Data from all the locations (thermocouples) is compared against each other to verify them for each test. Trends are similar to SOC 100 tests, however, TR max temperature reaches higher in this case.

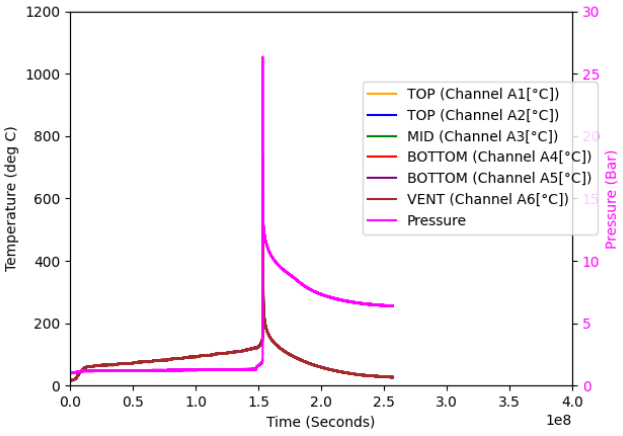


Figure 13: Test 3; Temperature and Pressure vs Time

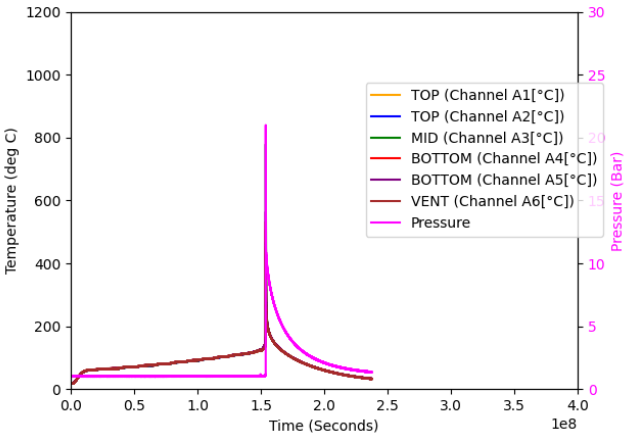


Figure 12: Test 4; Temperature and Pressure vs Time

SOC 50%:

Two tests; test number 5, 6 were carries out with SOC level 50%. Data from all the locations (thermocouples) is compared against each other to verify them for each test. Trends are similar to SOC 90% tests. lowest SOC in the tests that underwent thermal runaway reaction

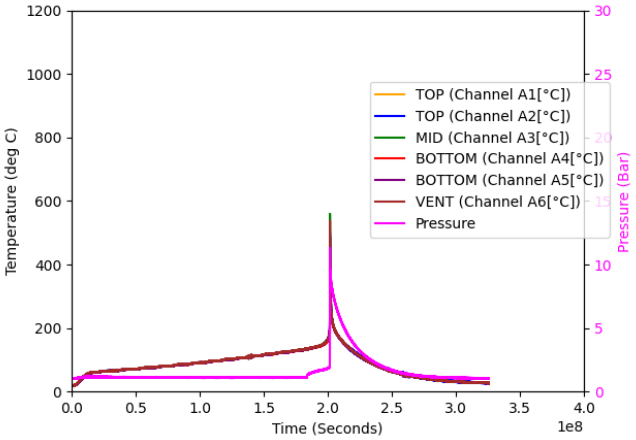


Figure 15 Test 5; Temperature and Pressure vs Time

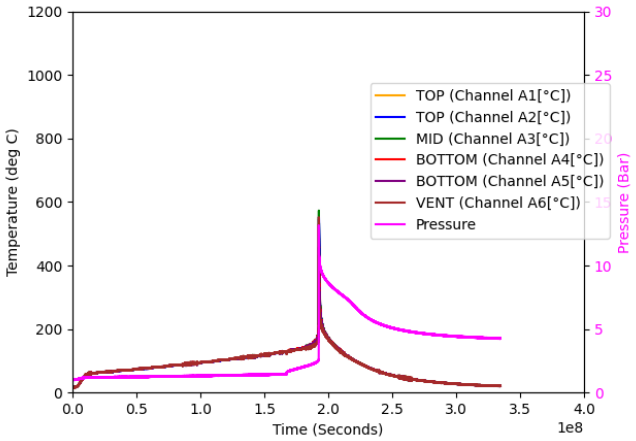


Figure 14: Test 6; Temperature and Pressure vs Time

SOC 30%:

Two tests; test number 7, 11 were carries out with SOC level 30%. Data from all the locations (thermocouples) is compared against each other to verify them for each test. Doesnot undergo TR.

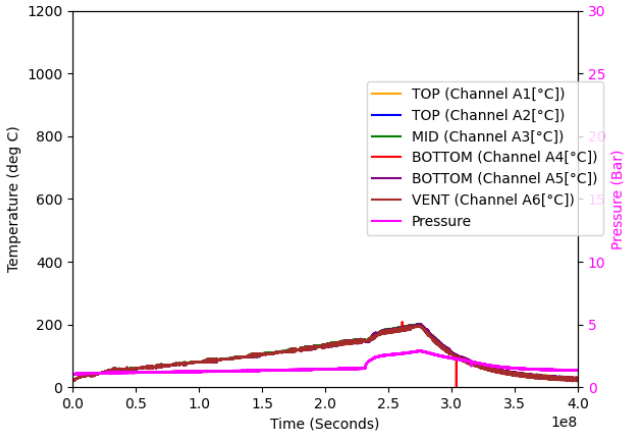


Figure 17: Test 7; Temperature and Pressure vs Time

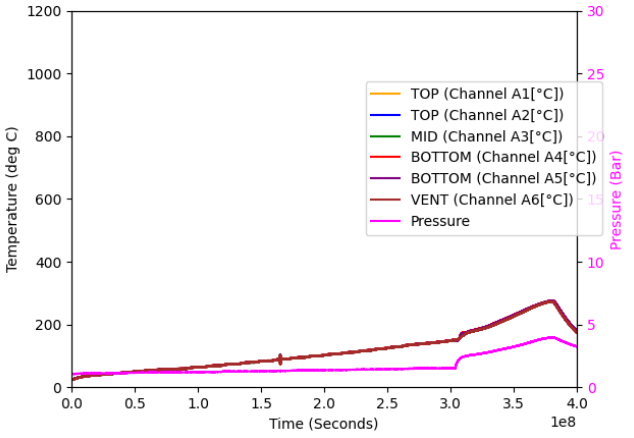


Figure 16: Test 11; Temperature and Pressure vs Time

SOC 07%:

Two tests; test number 8, 10 were carries out with SOC level 07%. Data from all the locations (thermocouples) is compared against each other to verify them for each test. Doesnot undergo TR

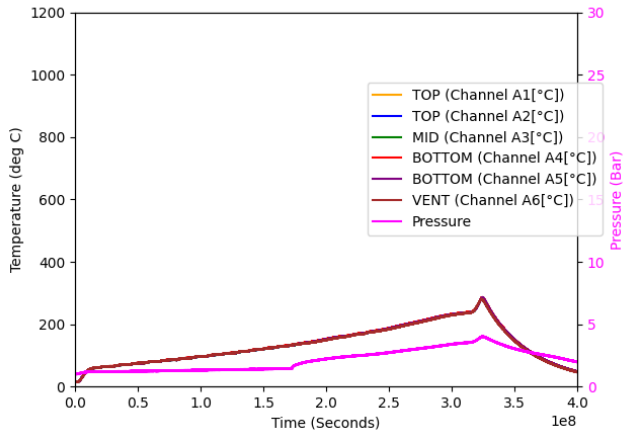


Figure 19: Test 8; Temperature and Pressure vs Time

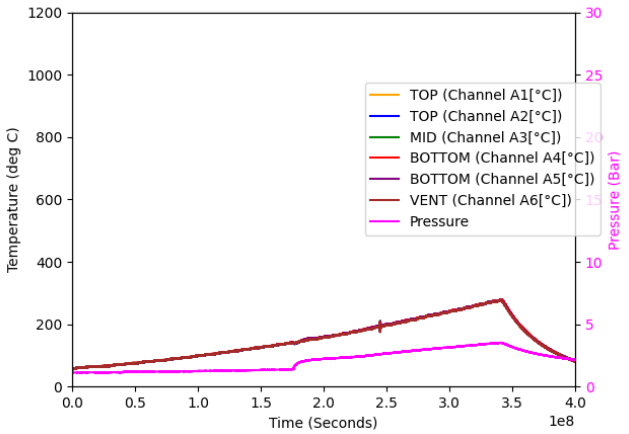


Figure 18: Test 10; Temperature and Pressure vs Time

After the intra cell analysis, inter-cell temperature comparison is carried out with time as the other parameter. This analysis is carried out to understand how temperature profile changes with respect to time and SOC level. Middle location of the Cell is used for the comparison here, reason being on the large, the middle location of cell was consistent across various tests with same SOC.

It can be clearly understood from this that as SOC decreases, time required to achieve TR increases, albeit not linearly. It can also be seen that TR is more intense at SOC 90%, rather than SOC 100%.

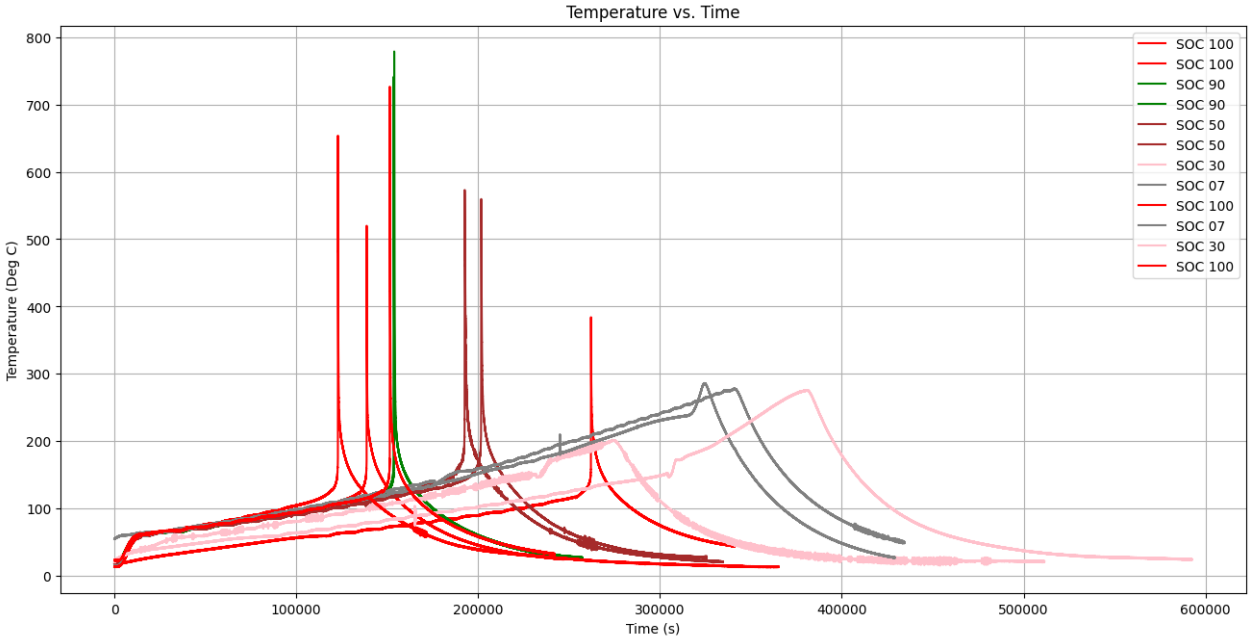


Figure 20: Temperature vs Time for different SOC

4.2 Temperature Rate analysis

After temperature analysis is carried out, Temperature rate is another important parameter that helps determine the TR intensity. Temperature rate is also used to define the process of self-heating. At any point of time, if cell has a steady rise in temperature at the rate of 0.02 deg C per minute, the cell assumed to be in self – heating stage of TR. Since only cells at SOC 50%, 90% and 100% undergo TR, only they are analyzed for this part of procedure.

Logarithmic Y axis is used to analyze the rate of change of temperature in order to reduce skewness and visualize relative changes in a better way. A lot of research papers have adopted significant model to represent rate of change of temperature in order to accommodate the spikes in the curve, due to explosive nature of reaction[7].

SOC 100%:

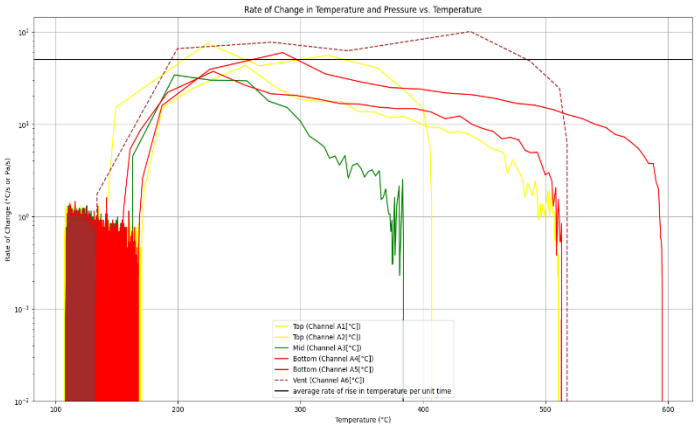


Figure 21: Test 1; Temperature Rate vs Time

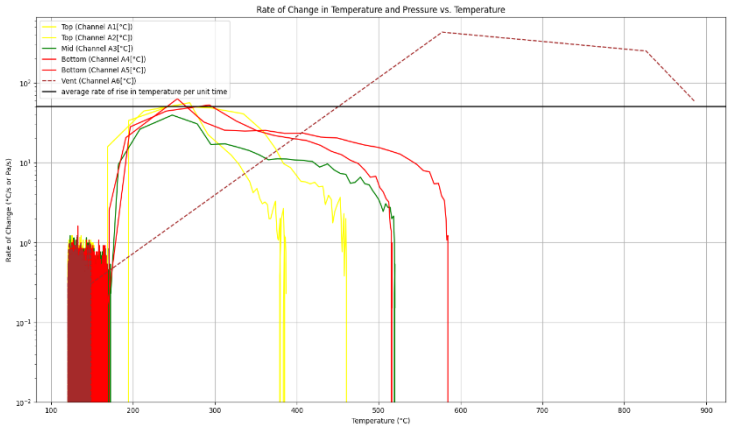


Figure 22: Test 2; Temperature Rate vs Time

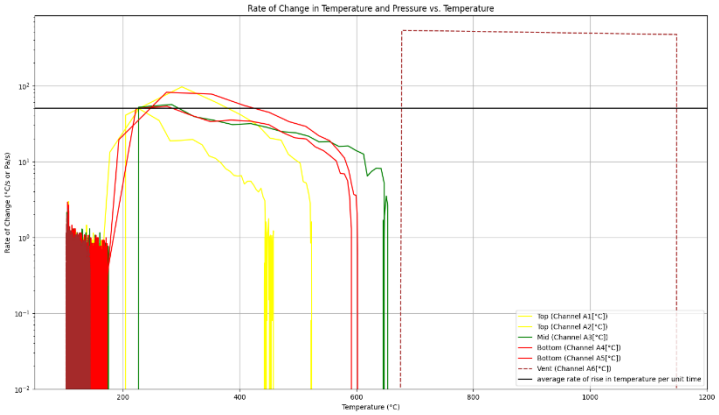


Figure 24: Test 9; Temperature Rate vs Time

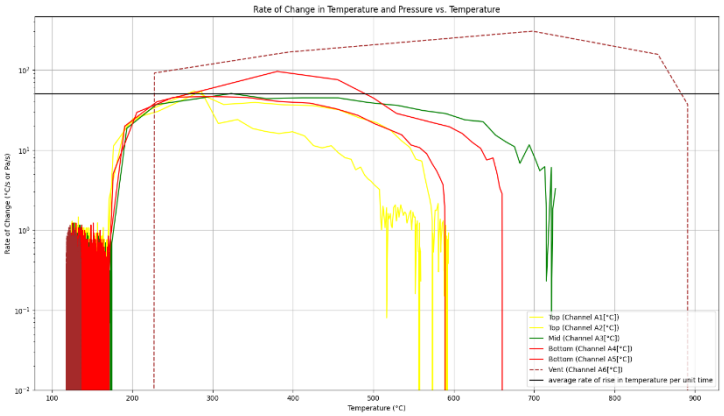


Figure 23: Test 12; Temperature Rate vs Time

A reference line at Rate of change of Temperature = 50 deg C per second. All the values are similar for maximum rate of change of temperature for all the locations across the tests. However middle location shows the most consistent values over the 4 tests for SOC 100%. Other key observation is thermocouple at vent experiences high intensity consistently.

SOC 90%

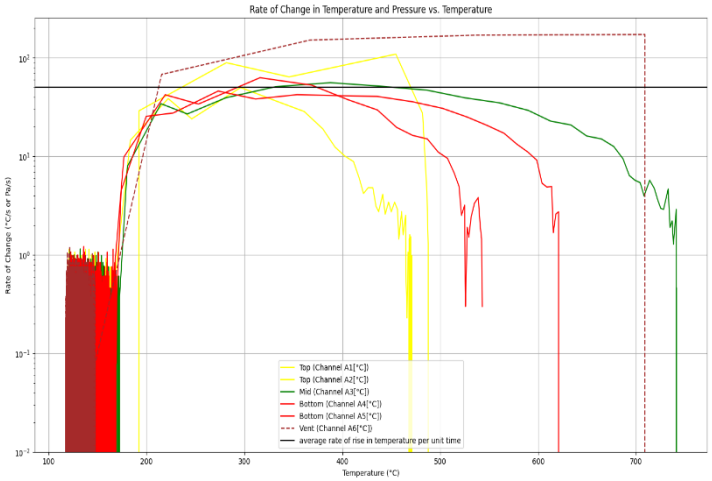


Figure 26: Test 3; Temperature Rate vs Time

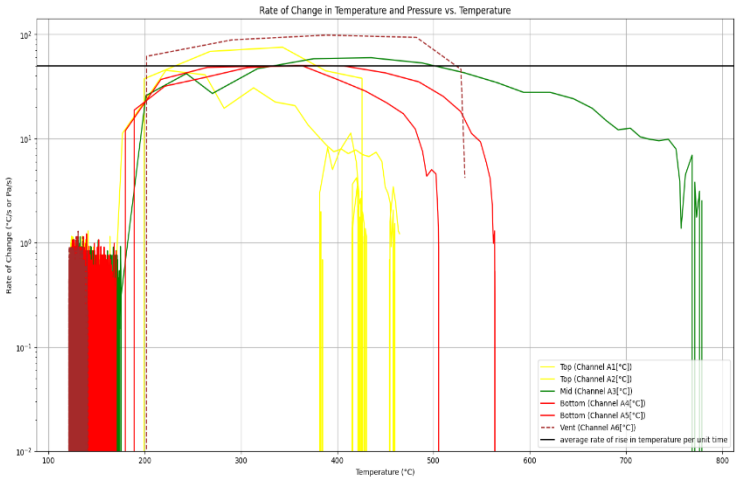


Figure 25: Test 4; Temperature Rate vs Time

Trends are similar to SOC 100% with maximum temperature rate change average around 50 deg Celsius per second. Mid location once again is consistent throughout the tests in comparison to other locations. Other key observation is thermocouple at vent experiences high intensity consistently, however, not as high as in the case of SOC 100%.

SOC 50%

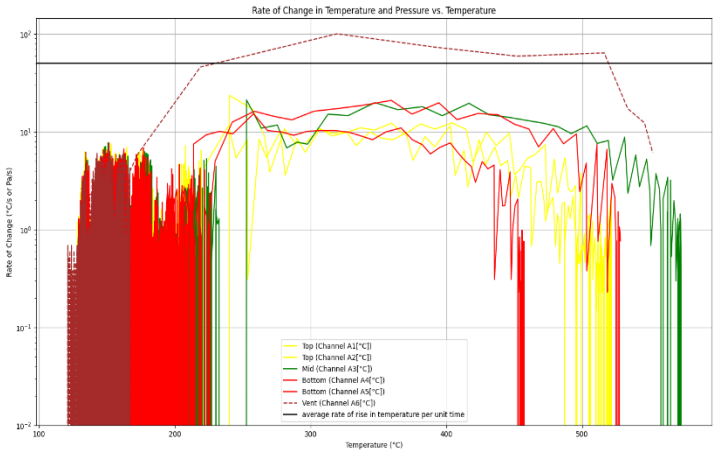


Figure 28: Test 5; Temperature Rate vs Time

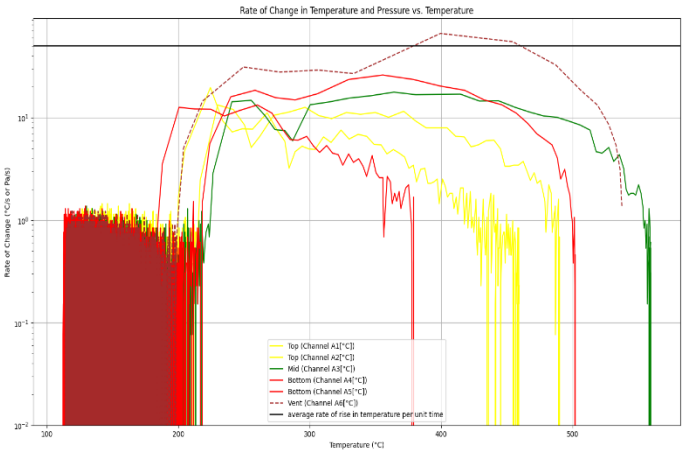


Figure 27: Test 6; Temperature Rate vs Time

While the general trends are similar to SOC 100% and 90%, the maximum temperature rate change average is significantly 50 deg C per second. Mid location once again is consistent throughout the tests in comparison to other locations. Other key observation is thermocouple at vent experiences high intensity consistently, however, not as high as in the case of SOC 100%.
Key trends comparing across all SOC levels:

- Vent Thermocouple consistently recorded highest temperature rise rates due to rapid gas release and exothermic reactions.
- Mid-section Thermocouple showed steady temperature increase over time, indicating it as the most thermally active zone.
- Temperature profiles were inconsistent across cell position at lower SOC levels, but at higher SOCs, the bottom section exhibited higher maximum temperature.

4.3 Maximum Temperature Analysis

All the previous results indicate the middle section to have most consistent results all across the different tests. Hence critical temperature points across all the tests on mid location are compared with each other. Various critical temperature points compared are :

- Maximum Temperature
- TR initiation Temperature
- Venting Temperature
- Self – heating Temperature

After analysing and plotting all the curves for temperature vs time for each experiment, all the above mentioned parameters are then found from the curve. those parameters are tabulated below

Table 1: Maximum temperature for all the locations at their respective SOC level

Max Temp for TR

SOC%	**Top 1**	**Top 2**	**Mid 1**	**Bottom 1**	**Bottom 2**	**Vent**
7.00	285.11	284.35	286.10	286.71	285.57	280.57
7.00	277.94	277.63	278.17	279.16	277.79	277.02
30.00	198.90	199.13	201.80	208.28	200.27	196.00
30.00	275.96	276.18	275.80	276.34	275.96	273.59
50.00	459.44	490.03	559.54	379.18	502.09	537.49
50.00	522.92	553.74	572.97	528.34	457.23	552.06
90.00	487.67	470.73	741.12	620.80	542.98	692.50
90.00	431.00	465.00	779.00	505.00	564.00	534.00
100.00	406.95	510.64	383.91	595.09	513.08	517.50
100.00	386.89	460.51	519.79	584.49	516.28	885.47
100.00	523.45	457.69	653.76	601.81	591.43	1148.50
100.00	594.48	561.07	726.55	588.99	660.32	891.04

Table 2: Self heating, Venting and Initiation temperature for all Locations at their respective SOC level

SOC%	**Temperature**		**Initation Temperature for TR**					
	Self H	**Vent**	**Top 1**	**Top 2**	**Mid 1**	**Bottom 1**	**Bottom 2**	**Vent**
7.00	134.15	134.15	-	-	-	-	-	-
7.00	140.5	140.5	-	-	-	-	-	-
30.00	140	150	-	-	-	-	-	-
30.00	133.4	151	-	-	-	-	-	-
50.00	130	134.5	198	214	221	183.1	217	198
50.00	126	129.88	219	219	230	223	207	162
90.00	111	121	163.7	167.8	172	169	166.4	147.1
90.00	110	122	161.00	164.00	170.10	165.40	168.90	155.40
100.00	109	116	131	170	158	168	155	133
100.00	110	123	152	160	173	168	168	148
100.00	105	126.5	163	163	173	172.3	171.5	143
100.00	108.7	125	168	164.5	173.2	170.2	170.3	135.5

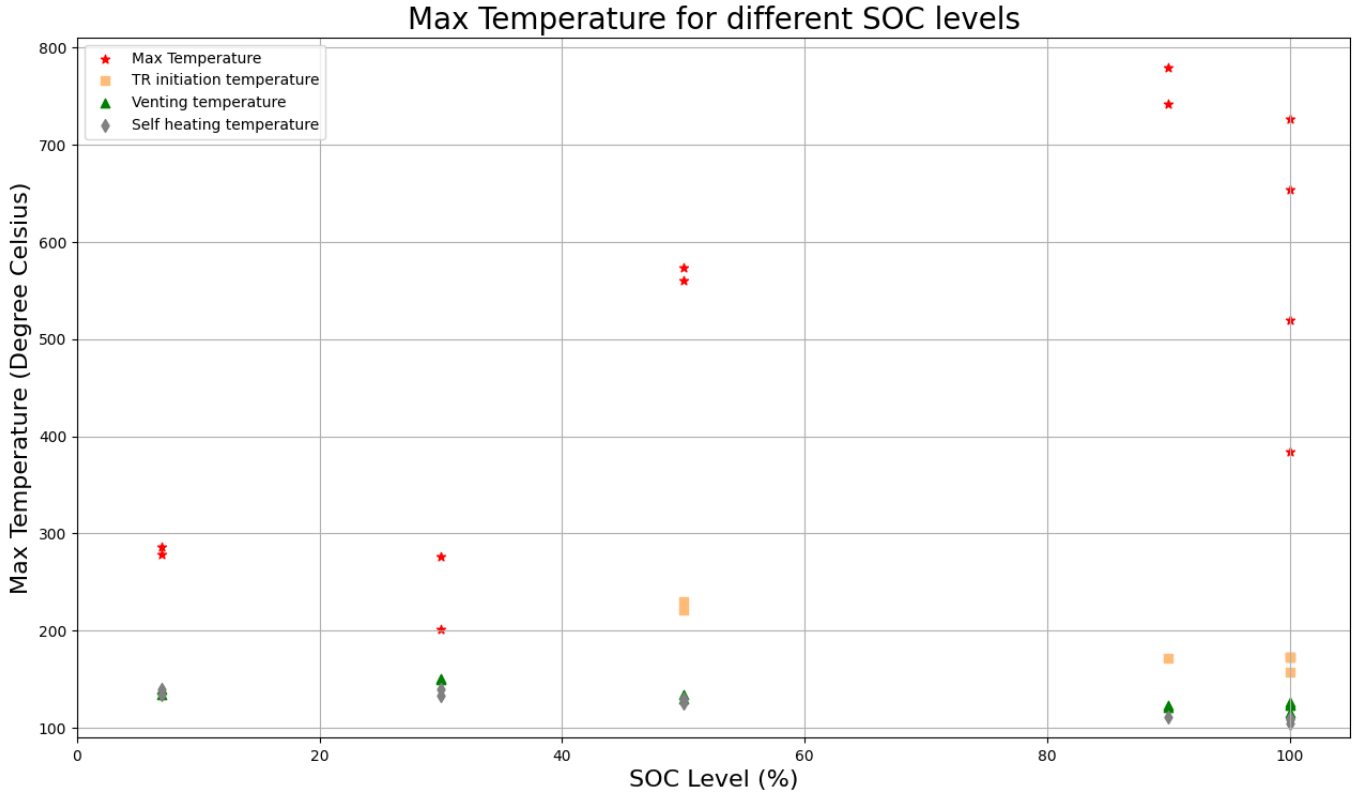


Figure 29: Max Temperature vs SOC

A dip in maximum temperature was observed around SOC 90%. TR initiation temperature and self-heating temperature showed a consistent decline across increasing SOC levels. This implies intensity of Thermal runaway is not just determined by maximum temperature and SOC level, hence other parameters need to be analyzed as well.

4.4 Multiple Parameter analysis

To assess TR intensity, parameters beyond temperature were analyzed. Max temperature, Max pressure and Mass loss fraction were analyzed with respect to SOC. Mass Loss fraction data was calculated by weighing the final mass after thermal runaway and calculating the lost fraction using the mass of cell before experiment

Table 3: Max Temperature at Mid section in comparison the Max Pressure and Mass loss Fraction for all the SOC levels

SOC%	Max T (Celsius)	Max P (bar)	Mass loss f
7.00	286.10	4	7.00
7.00	278.17	3.5	6.52
30.00	201.80	2.9	6.62
30.00	275.80	3.9	6.35
50.00	559.54	11.3	50.89
50.00	572.97	13.1	43.83
90.00	741.12	26.3	63.40
90.00	779.00	18.5	64.05
100.00	383.91	27.7	66.85
100.00	519.79	28.6	72.89
100.00	653.76	28.9	68.44
100.00	726.55	28.8	64.00

Max Pressure and Mass Loss Fraction showed distinct trends from temperature, especially at higher SOCs. Unlike Temperature, Max pressure and Mass loss fraction did not show any dip around SOC level 90%

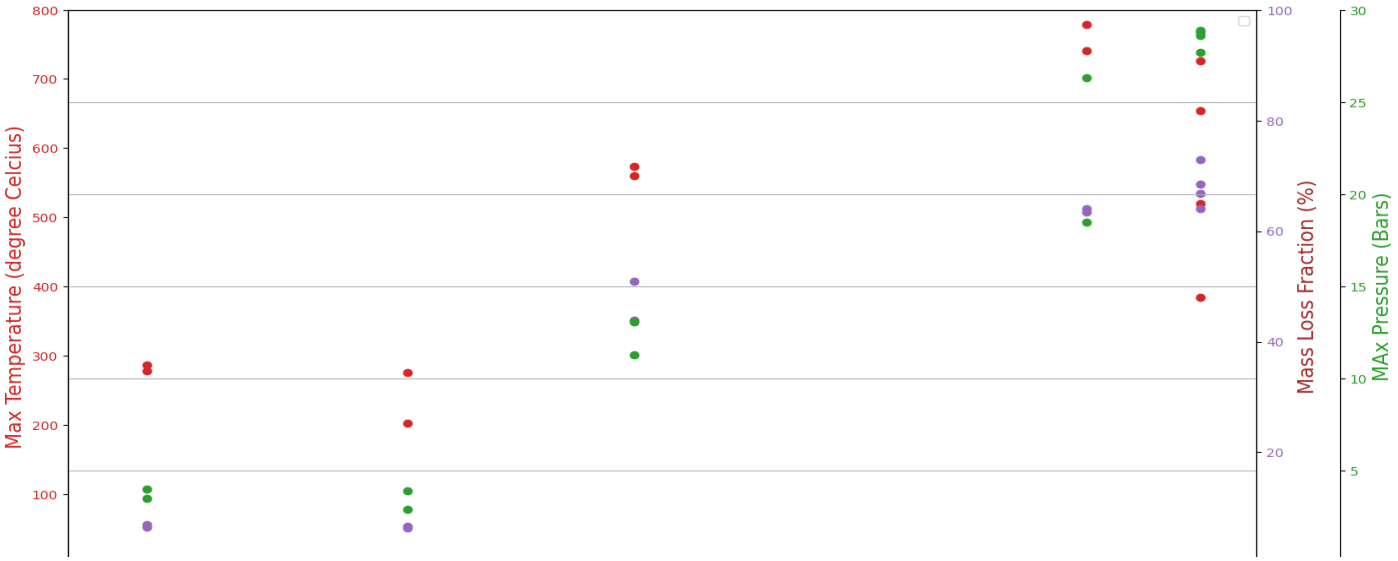


Figure 30: Multiple Parameter analysis (Temperature, Pressure and Mass Loss Fraction)

4.5 Mass Loss Fraction Analysis

Mass loss fraction analysis is carried out with respect to max temperature and max pressure to study the dependence of temperature and pressure on mass loss fraction

Mass loss fraction vs Max Pressure

Steep positive correlation was observed, indicating gas release increases exponentially with mass loss fraction increase.

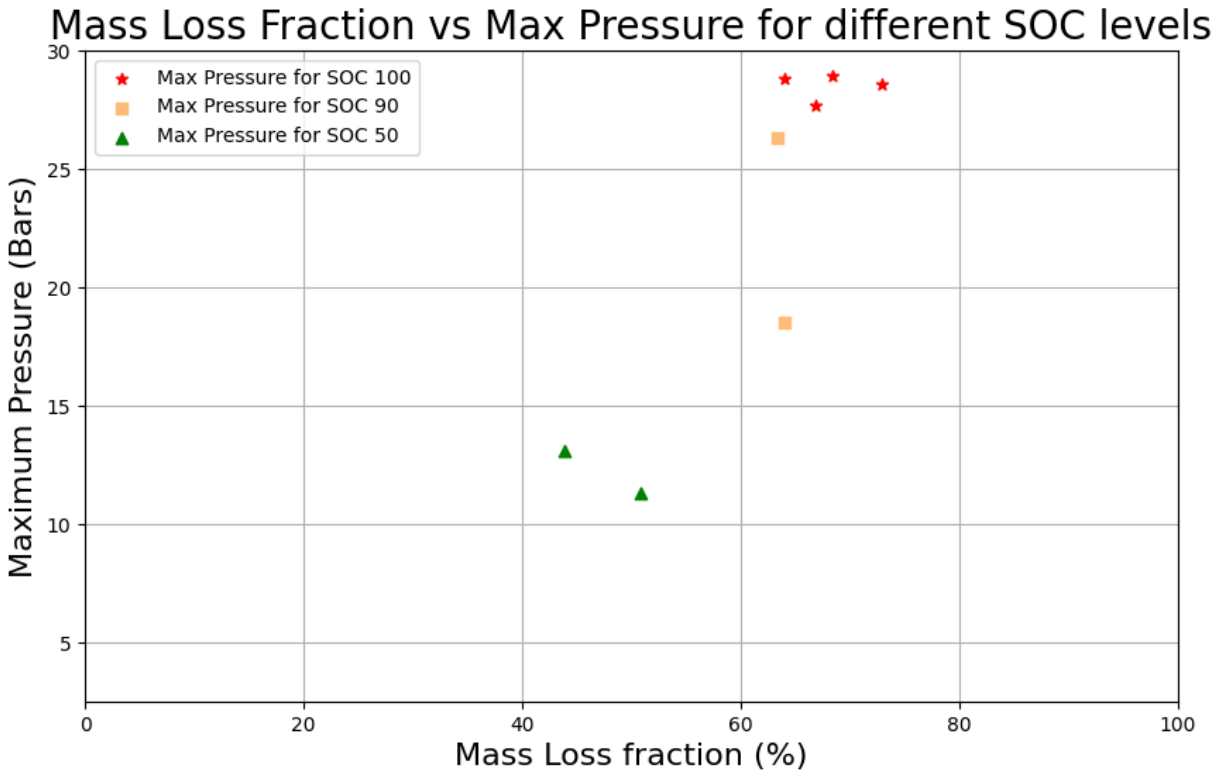


Figure 31: MLF vs Max Pressure at different SOC

Mass loss fraction vs Max Temperature

An inverse relationship was consistently observed at **SAME** SOC Level. Intense thermal events lead to violent venting and higher mass loss. This mass loss in turn carries the heat away from cell, resulting in lower temperature detection at the thermocouple. It can be observed via Fig 32-35, the trend for mass loss fraction and max temperature stay consistent throughout the different locations on the experiment cell.

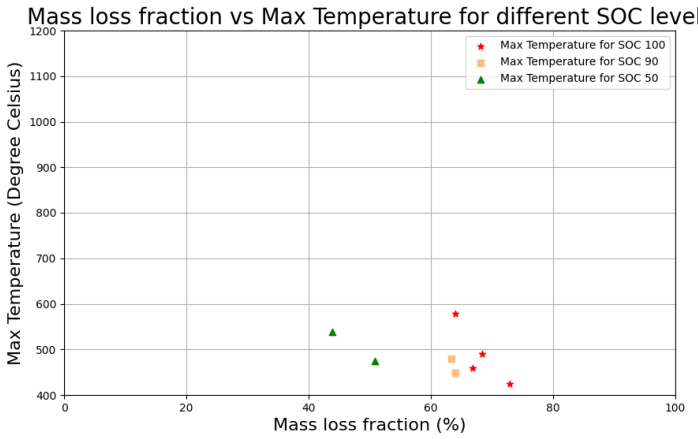


Figure 35: MLF vs Max Temperature for different SOC for Top location

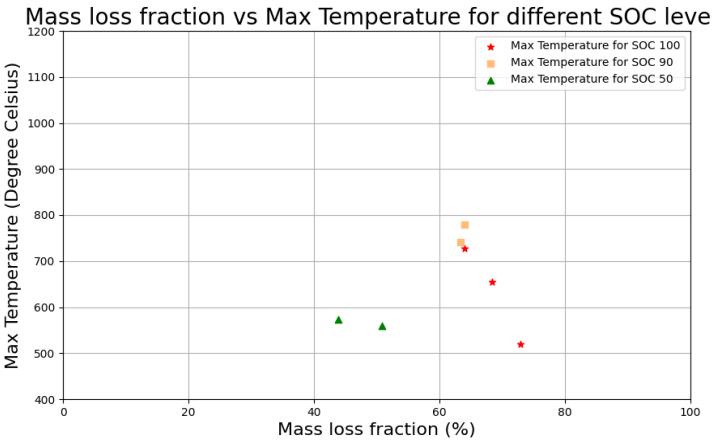


Figure 34: MLF vs Max Temperature for different SOC for Mid location

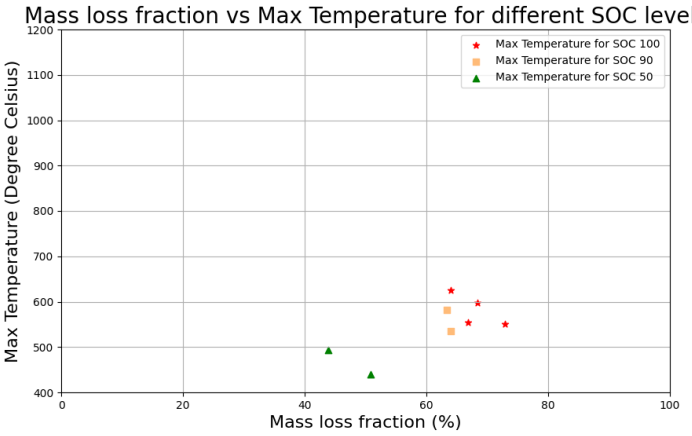


Figure 33: MLF vs Max Temperature for different SOC for Bottom location

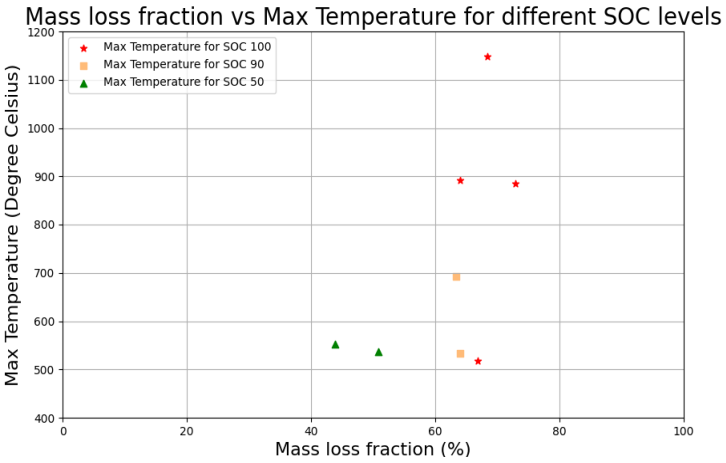


Figure 32: MLF vs Max Temperature for different SOC for Vent location

4.6 Venting Pressure Analysis

An unusual trend was noticed when venting pressure and maximum temperature were analyzed together.

At higher SOCs, venting pressure significantly influenced the maximum temperature achieved. But at lower SOC level, venting pressure did not contribute to major changes in maximum temperature. Since venting pressure is analyzed, the Vent, which measure the temperature around vent is used to for analysis. This implies the vent can play a major role in cell safety if pressure is controlled appropriately.

This correlations is explained by electrolyte decomposition and exothermic reactions which generate gases like CO_2, CO and ethene.[11]

Table 4: Venting Pressure and Vent Max temperatures at various SOC

Test	SOC%	Vent	Vent
08	7.00	280.57	
10	7.00	277.02	1.49
07	30.00	196.00	1.7
11	30.00	273.59	1.68
05	50.00	537.49	1.25
06	50.00	552.06	1.67
03	90.00	692.50	1.425
04	90.00	534.00	1.15
01	100.00	517.50	1.16
02	100.00	885.47	1.44
09	100.00	1148.50	1.47
12	100.00	891.04	1.4

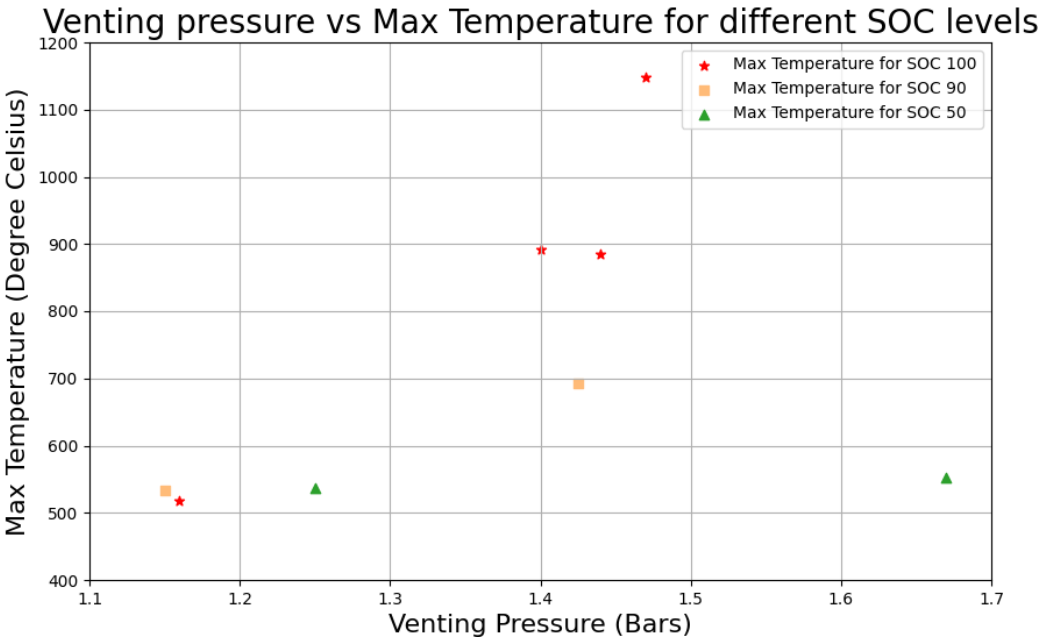


Figure 36: Venting Pressure vs Max Temperature at different SOC levels

5. CONCLUSIONS

A discussion of the results and the conclusions drawn during the thesis are presented in this chapter. The conclusions are based from the analysis with the intention to answer the formulation of questions that is presented in Chapter 1.

Various thermal thresholds associated with thermal runaway such as Self heating, TR Initiation and TR max temperature where calculated. It was observed that cell will not undergo TR no matter how high the temperature goes if the SOC level is too low. Critical SOC level for TR was identified to be 50%.

Table 5: Range of Temperature and pressure over SOC level

SOC%	T. Range (deg C)	P. Range (Bar)
50	487-531	011-013
90	545-592	18-26
100	487-670	27-29

The ARC testing of NMC811 cells revealed that thermocouple placement and SOC levels significantly influence thermal runaway detection. Based on the maximum temperature and temperature rate analysis, it was observed that mid-section of battery stood out among all other section in terms of consistent temperatures for similar SOC level. For vent section, the thermocouple at vent consistently provided higher temperature than rest of the sections, but also provided essential insights. Hence the optimum position to measure a cells temperature during TR is mid-section followed by vent section.

Mass loss fraction and venting pressure emerged as key indicators of TR severity, with inverse relationships observed between mass loss and temperature. High mass loss indicates more gas production and an intense TR, as it can be associated with high max pressure from Fig. 31. This indicates higher heat production as well since the gas generation reactions are usually exothermic. It is demonstrated that mass loss, especially at high SOC level promotes heat emission which go unnoticed if measured via cell temperature.

These findings underscore the importance of sensor positioning and vent design in enhancing battery safety. The study successfully met its objectives by quantifying thermal thresholds, identifying failure sequences, and evaluating safety features. Venting mechanism proved to be highly effective in increasing the safety of the cell. early venting can lead to lower temperatures and less reactions inside cell.

6 RECOMMENDATIONS AND FUTURE WORK

In this chapter, recommendations on more detailed solutions and/or future work in this field are presented.

6.1 Recommendations

Based on the findings, it is recommended to optimize thermocouple placement in abuse testing setups, particularly at the mid-section and vent locations. Battery manufacturers should consider reinforcing venting mechanisms to manage pressure buildup effectively. Standardized preconditioning of cells before testing is essential to ensure consistent thermal behavior. Instrumentation improvements, such as robust pressure lines and high-resolution sensors, are also advised for accurate data capture.

One of the practical shortcoming while carrying out experiment was suboptimal cell support at times inside the bomb calorimeter. This can lead to loosening up of the thermocouple which in turn will lead to noisy and inaccurate data. Vent opening up on one side of the cell induces a bias that was not considered, coming up with a way to avoid such bias would further improve the accuracy of result.

6.2 Future Work

Future research should explore the behavior of multi-cell modules to understand inter-cell interactions during thermal runaway. Expanding the study to include other chemistries such as LFP and NCA would provide comparative insights. Advanced gas analysis techniques could be employed to characterize the composition of vented gases. Additionally, real-world abuse scenarios like mechanical impact and electrical overcharge should be simulated to validate laboratory findings

To further build up on this project, similar thermal runaway tests should be carried out with a way to control the venting pressure. This will help confirm the venting pressure relation with thermal runaway intensity. Gas analysis of the emissions after thermal runaway can help understand the reactions in the cell and if there is a significant different in composition of gas in the case with early venting vs late venting. If done right, this can even help identify the active material in the reaction.

Based on the temperature data and chemical composition of battery, a kinetic model that simulated the thermal runaway can be modeled in order to understand the thermal runaway in a better way.

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